

Two-Column Congruence Proofs Involving Triangles

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A **two-column proof** is an organized way of proving that a mathematical statement is true.

Statements and Reasons

The left column contains the **statements**, while the right column contains the corresponding **reasons**.

Statements	Reasons
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Every statement must have a valid mathematical reason.

Writing a Two-Column Proof

Begin with the **given information**.

Then use definitions, postulates, and theorems to justify each step.

Given $\rightarrow$ Statements $\rightarrow$ Conclusion

Triangle Congruence

Common triangle congruence postulates are

SSS

SAS

ASA

AAS

and the **HL Theorem** for right triangles.

Using CPCTC

Once two triangles are proven congruent, use

CPCTC

Corresponding Parts of Congruent Triangles are Congruent.

This proves corresponding sides or angles are congruent.

Logical Sequence

Every statement in a proof must have a valid mathematical reason.

A proof should follow a logical sequence from the given information to the desired conclusion.

Examples

Example 1

Given

$$AB \cong DE, \quad BC \cong EF, \quad \angle B \cong \angle E$$

Prove

$$\triangle ABC \cong \triangle DEF.$$

The congruence postulate is

SAS

Example 1: Proof

Statements	Reasons
$AB \cong DE$	Given
$BC \cong EF$	Given
$\angle B \cong \angle E$	Given
$\triangle ABC \cong \triangle DEF$	SAS

Example 2

Given

$$AB \cong DE, \quad AC \cong DF, \quad BC \cong EF$$

Prove

$$\triangle ABC \cong \triangle DEF.$$

The congruence postulate is

SSS

Example 2: Proof

Statements	Reasons
$AB \cong DE$	Given
$AC \cong DF$	Given
$BC \cong EF$	Given
$\triangle ABC \cong \triangle DEF$	SSS

Example 3

Given

$$\triangle ABC \cong \triangle DEF$$

Prove

$$\angle C \cong \angle F.$$

Use CPCTC.

Example 3: Proof

Statements	Reasons
$\triangle ABC \cong \triangle DEF$	Given
$\angle C \cong \angle F$	CPCTC

Example 4

Given

$$AB \cong CD, \quad \angle A \cong \angle C, \quad \angle B \cong \angle D$$

Prove

$$\triangle ABE \cong \triangle CDF.$$

The congruence postulate is

ASA

Example 4: Proof

Statements	Reasons
$AB \cong CD$	Given
$\angle A \cong \angle C$	Given
$\angle B \cong \angle D$	Given
$\triangle ABE \cong \triangle CDF$	ASA

Example 5

Given

$$\triangle ABC \cong \triangle DEF$$

Prove

$$\overline{AC} \cong \overline{DF}.$$

Use CPCTC.

Example 5: Proof

Statements	Reasons
$\triangle ABC \cong \triangle DEF$	Given
$\overline{AC} \cong \overline{DF}$	CPCTC

Practice Exercises

Practice 1

In $\triangle ABC$ and $\triangle DEF$,

$$AB \cong DE, \quad BC \cong EF, \quad \angle B \cong \angle E$$

Construct a two-column proof that

$$\triangle ABC \cong \triangle DEF.$$

Practice 2

Given

$$AB \cong DE, \quad AC \cong DF, \quad BC \cong EF$$

Construct a two-column proof that

$$\triangle ABC \cong \triangle DEF.$$

Practice 3

Given

$$\triangle ABC \cong \triangle DEF$$

Construct a two-column proof that

$$\angle C \cong \angle F.$$

Practice 4

Given

$$AB \cong CD, \quad \angle A \cong \angle C, \quad \angle B \cong \angle D$$

Construct a two-column proof that

$$\triangle ABE \cong \triangle CDF.$$

Practice 5

Given

$$\triangle ABC \cong \triangle DEF$$

Construct a two-column proof that

$$\overline{AC} \cong \overline{DF}.$$

Activities

Activity 1

Given

$$\angle M \cong \angle P, \quad \angle N \cong \angle Q, \quad MN \cong PQ$$

Construct a two-column proof that

$$\triangle MNO \cong \triangle PQR.$$

Activity 2

In right triangles $\triangle RST$ and $\triangle ABC$,

$$\angle S, \angle B = 90^\circ$$

$$RT \cong AC, \quad ST \cong BC$$

Construct a two-column proof that the triangles are congruent.

Activity 3

Given

$$\triangle PQR \cong \triangle XYZ$$

Construct a two-column proof that

$$QR \cong YZ.$$

Activity 4

In $\triangle RST$, RS is the common side of $\triangle RSM$ and $\triangle RSN$.

Given

$$RM \cong RN, \quad SM \cong SN$$

Construct a two-column proof that

$$\triangle RSM \cong \triangle RSN.$$

Activity 5

Given

$$LM \parallel NO, \quad \angle L \cong \angle N, \quad MN \cong LO$$

Construct a two-column proof that

$$\triangle LMN \cong \triangle NOL.$$

Thank you!